

Companion to: *A Band-Mask Robustness Diagnostic for LDA on HSI*

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This supplement expands Section V of the conference paper.

Mechanism (ii): weak identification. KSC has 13 classes, $D = 2686$ in the canonical sample, and $B = 176$, giving a posterior over ϕ with ~ 2100 parameters and ~ 2700 documents. The posterior has substantial mass over many high-likelihood configurations, and a band restriction selects a different posterior mode of the same family. The masked fit is not *wrong*, it is merely a different sample from a weakly informative posterior.

Discriminating between (i) and (ii) is out of scope of the diagnostic. A natural follow-up is a posterior-mode-counting diagnostic (refit under many seeds and cluster the resulting $\{\phi^{(s)}\}$ for $s = 1, \dots, N$ seeds; count distinct clusters above a cosine-similarity threshold). The companion code repository already ships such a diagnostic under `topic_stability/` and `deep_seed_stability/` but its interpretation is outside this conference paper’s scope.

D.4 Reading the ARI-vs-label column

A subtle but important point. The ARI-vs-label column in Suppl D Table I reports ARI(masked dominant topic, ground truth). A high value means the masked fit is still discriminating the ground-truth classes even though its topic identities have changed. The clearest example is Pavia U $\times$ VNIR with paired ARI = 0.415 and ARI-vs-label = 0.316: the masked fit is still useful for downstream classification (it preserves about as much label discrimination as the canonical fit, ARI $\approx$ 0.32) even though the topic identities have shifted. Conversely, Botswana $\times$ SWIR has paired ARI = 0.013 *and* ARI-vs-label = 0.098: the masked fit is both topically unstable and operationally weaker.

E.5 What the diagnostic does NOT claim

We close with four explicit non-claims:

It does not claim a single best band selection. The top-50 Fisher mask achieves the lowest paired-ARI on Indian Pines (0.398) and the highest on Pavia U (0.387), so the same “50-band-budget” mask is interpretable differently on different scenes. Selecting a band budget for a downstream product is a separate optimisation that should marginalise over the per-scene diagnostic.

It does not claim a chance-baseline threshold. Statistical significance of paired ARI under a permutation null is not reported here; the threshold for “meaningful agreement” is a matter of judgement on the present sample sizes ($D = 1320\text{--}3520$). A formal permutation test is in the companion code repository under `validation_blocks/`.

It does not claim that band-fragility implies model inadequacy. A scene whose canonical topics are band-fragile may still admit a useful interpretation under a different inductive bias (e.g. deep autoencoder embedding); the diagnostic asks specifically whether the *LDA* basis is band-robust, not whether the data admits any band-robust representation.

It does not claim transferability to non-AVIRIS sensors without recalibration. The band-mask wavelengths are read from the per-scene `wavelengths_nm` metadata; running the diagnostic on a sensor with different band centres requires recomputing the mask membership functions but no algorithmic change.

References